A fiberwise transformation is a fiberwise equivalence exactly when its total map is an equivalence

The Univalent Foundations Program

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Editorial note. The counted statement is the author thm environment at source lines 767-771 and its proof environment opens on line 773, the line after the blank separator; the pairing is by adjacency and the retained proof range 773-780 is the whole environment.

Editorial note. The book label-number dump gives this label as 4.7.7 (env thm, chapter 4, section 7, seventh counter value), so the counters chapter 4, section 7 and thm 6 (zero-based) are set so that the item prints with the book number.

Editorial note. One label the proof cites is declared in external_references as a non-load-bearing navigation pointer: fibwise-fiber-total-fiber-equiv in this chapter, the comparison of the two kinds of homotopy fibre; it is counted by its own card of this lane and the retained proof carries out the regrouping that follows it.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the total map of a fiberwise transformation is an equivalence exactly when the transformation is fiberwise an equivalence (author label thm:total-fiber-equiv, book Theorem 4.7.7).

Allowed standard prerequisites (editorial).

- Homotopy fibres of the total map. For a fiberwise transformation between two families over a type, the homotopy fibre of the total map at a point is equivalent to the homotopy fibre of the fiberwise map at the second component, so contractibility transfers.. It is the pointwise input of the proof, cited by the retained text.
- Fiberwise equivalence as pointwise contractibility. A fiberwise transformation is a fiberwise equivalence exactly when each of its homotopy fibres is contractible, and a total map is an equivalence exactly when each of its homotopy fibres is contractible.. It converts the pointwise statement about contractible fibres into the equivalence of the two maps.

Source reference. Original lemma 4.7.6: sources/hott-equivalences/pinned-original-source.tex, source lines 736–736. Book Lemma 4.7.6, comparing the homotopy

fibre of the total map with the homotopy fibre of the fiberwise map, cited at the start of the retained proof. Navigation pointer only: it is counted by its own card of this lane.

Theorem 4.7.7. *Suppose that f is a fiberwise transformation between families P and Q over a type A. Then f is a fiberwise equivalence if and only if* $\mathsf{total}(f)$ *is an equivalence.*

Proof. Let f, P, Q and A be as in the statement of the theorem. By original lemma 4.7.6 it follows for all $x : A$ and $v : Q(x)$ that $\mathsf{fib}_{\mathsf{total}(f)}((x, v))$ is contractible if and only if $\mathsf{fib}_{f(x)}(v)$ is contractible. Thus, $\mathsf{fib}_{\mathsf{total}(f)}(w)$ is contractible for all $w : \sum_{(x:A)} Q(x)$ if and only if $\mathsf{fib}_{f(x)}(v)$ is contractible for all $x : A$ and $v : Q(x)$. □